Water Pollution



River Beas (Punjab)



River Cooum in Chennai (Tamil Nadu)

Contents

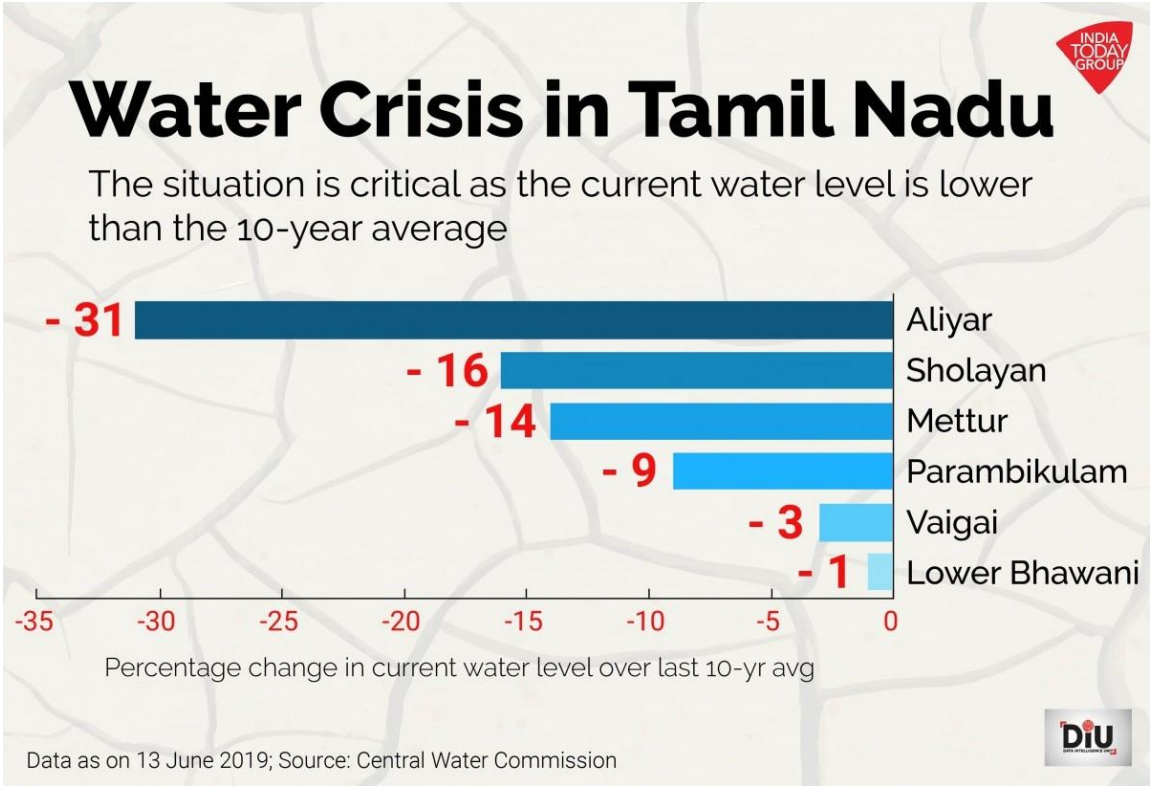


- Introduction
- Sources
- Impacts
- Treatment Methods

- **Water Pollution: Causes, Treatments and Solutions by Dr. Luxmy Begum**
- **Introduction to Water Pollution Biology by Richard J. Schmitz**
- **Elements of Water Pollution Control Engineering by O P Gupta**
- **https://cpcb.nic.in/**
- **https://cpcb.nic.in/nwmp-data-2021/**
- **http://www.cwc.gov.in/water-quality-inforamtion**
- **https://www.unep.org/explore-topics/water/what-we-do/tackling-global-water-pollution**

Introduction

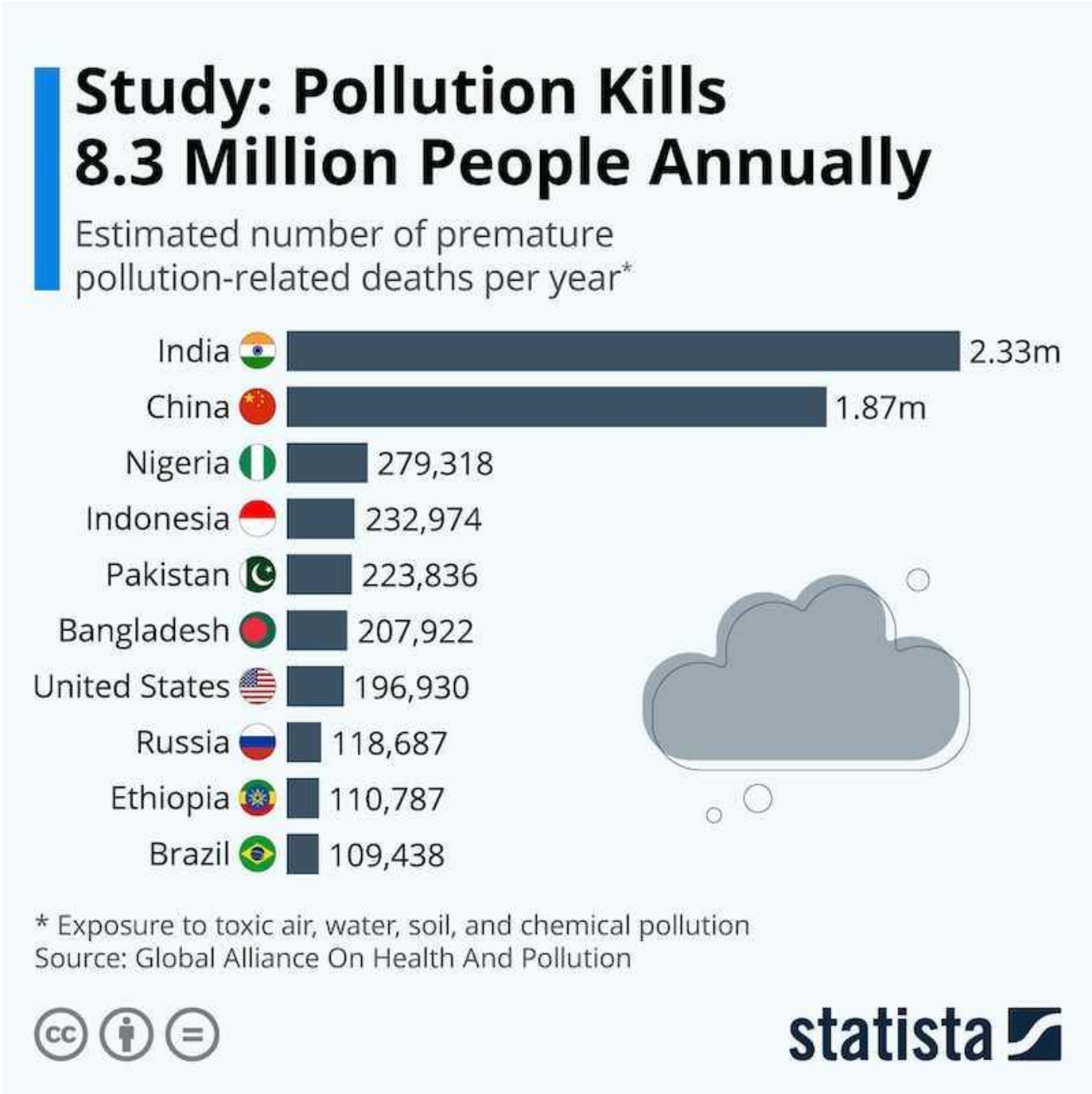
- Importance of water - necessary component of the life on earth.
- Water pollution is the contamination of water bodies
  - e.g. lakes, rivers, oceans and ground waters
- Water pollution occurs when pollutants are discharged directly or indirectly into water bodies without adequate treatment to remove harmful compounds.
- Some 80 percent of the world's wastewater is dumped largely untreated back into the environment, polluting rivers, lakes, and oceans.
- Water pollution affects plants and organisms living in these bodies of water.
- Drinkable water sources are finite: Less than 1 percent of the earth's freshwater is actually accessible to us.
- Potentially harmful contaminants—from arsenic to copper to lead—have been found in the tap water of every single state in the nation.



The National Institution for Transforming India (NITI Aayog), a thinktank by the Government of India reported that 600 million people in India encounter extreme to high water stress.

Introduction

- Some 829 000 people are estimated to die each year from diarrhoea as a result of unsafe drinking-water, sanitation and hand hygiene.
- Researchers surveyed more than 75,000 bodies of water in 89 countries and found that more than 40 per cent were severely polluted.
- An estimated 700 million Indians have no access to a proper toilet, and 1,000 Indian children die of diarrheal sickness every day.



Introduction

- More than 38,000 million liters of waste water enter Indian rivers due to lack of sewage treatment plants and improper waste disposal mechanisms.
- Over half of India's 605 rivers were found to be polluted by the Central Pollution Control Board in 2022.
- Municipal solid waste from India contributed to 10% of waste leakage to the world's rivers in 2020, according to a new study.







Ram Vilas Paswan
@irvpaswan · **Follow**

Most of the samples from almost all cities have failed to comply with the requirements of IS 10500:2012 in one or more parameters. Only samples drawn from Mumbai were found fit on all the parameters. 3/4 @PMOIndia @consaff @IndianStandards

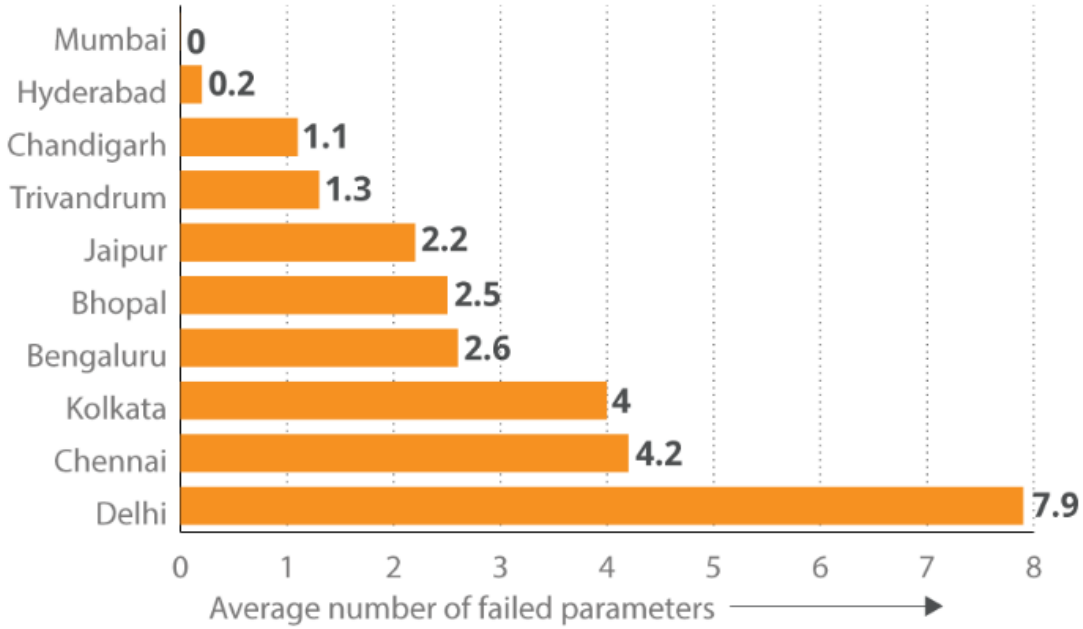
Categorisation	Capital	No. of Samples Failing	No. of Individual Parameters in which samples failing
1	Mumbai	0/10	0
2	Hyderabad	1/10	1
	Bhubaneswar	1/10	1
3	Ranchi	1/10	4
4	Raipur	5/10	3
5	Amravati	6/10	7
6	Shimla	9/10	1
7	Chandigarh	10/10	2
8	Thiruvananthapuram	10/10	3
9	Patna	10/10	4
	Bhopal	10/10	4
10	Guwahati	10/10	5
	Bengaluru	10/10	5
	Gandhinagar	10/10	5
11	Lucknow	10/10	6
	Jammu	10/10	6
12	Jaipur	10/10	7
	Dehradun	10/10	7
13	Chennai	10/10	9
14	Kolkata	9/9	10
15	Delhi	11//11	19



8:55 PM · Nov 16, 2019

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Average number of parameters on which cities' water failed tests



Types of Water Pollution

- Groundwater
  - When rain falls and seeps deep into the earth, filling the cracks, crevices, and porous spaces of an aquifer (basically an underground storehouse of water), it becomes groundwater—one of our least visible but most important natural resources.
  - When chemicals on the surface soil seep into the ground via rainwater.
  - This occurs most frequently in areas near farms, due to the use of chemical fertilizer.
  - Once polluted, an aquifer may be unusable for decades, or even thousands of years.

Types of Water Pollution

- Surface water
  - Covering about 70 percent of the earth, surface water is what fills our oceans, lakes, rivers, and all those other blue bits on the world map.
  - Includes lagoons, rivers, oceans, and lakes
  - Results from mixing surface water with pollutants like spilled oil or improperly disposed industrial waste.
  - There’s also all the random junk that industry and individuals dump directly into waterways.

Sources

- Point
  - When contamination originates from a single source, it's called point source pollution.
  - Examples include wastewater (also called effluent) discharged legally or illegally by a manufacturer, oil refinery, or wastewater treatment facility, as well as contamination from leaking septic systems, chemical and oil spills, and illegal dumping.

- Nonpoint
  - The identification of certain pollution sources is difficult such as agricultural run-offs, stormwater drainage, acid rain, etc.
  - It is difficult to recognize them because it is possible to have many agricultural runoffs like farms, animals or crop lands from where water can enter the water source.

Major pollutant and sources

Pollutant	**Source**
Organic wastes	Domestic sewage, decay of animals and plants, animal waste and, waste and discharge from food processing factories
Pesticides	Chemicals required to kill insects, weeds, fungi etc.
Microorganisms	Domestic sewage and discharges
Heat	Water discharges after undergoing cooling in industries
Toxic heavy metals	Different factories and plants such as chemical factory
Chemical fertilizers	Plant nutrients
Radioactive substances	Mining of Uranium/Thorium containing minerals
Sediments	Soil erosion due to strip mining and agriculture

Effects of water pollution

- Effects on aquatic ecosystem:
  - Polluted water reduces Dissolved Oxygen (DO) content, thereby, eliminates sensitive organisms like plankton, molluscs and fish etc.
  - Biocides, polychlorinated biphenyls (PCBs) and heavy metals directly eliminate sensitive aquatic organisms.
  - Hot waters discharged from industries, when added to water bodies, lowers its DO content.
- Effects on human health:
  - The polluted water usually contains pathogens like virus, bacteria, parasitic protozoa and worms; therefore, it is a source of water borne diseases like jaundice, cholera, typhoid, arnoebiasis etc.

Contamination of Water by Pathogens

- Domestic sewage and animal excreta contain different types of pathogens such as bacteria and other microorganisms. For instance, human excreta contain bacteria like Streptococcus faecalis and E. coli responsible for gastrointestinal diseases.
- High levels of pathogens may result from inadequately treated sewage discharges.
- In developed countries, older cities with aging infrastructure may have leaky sewage collection systems (pipes, pumps, valves), which can cause sanitary sewer overflows.
- Microorganisms sometimes found in surface waters which have caused human health problems include:
  - Burkholderia pseudomallei
  - Cryptosporidium
  - Giardia lamblia
  - Salmonella
  - Norovirus and other viruses
  - Parasitic worms (helminths)

Effects on human health

- Mercury compounds in waste water are converted by bacterial action into extremely toxic methyl mercury, which can cause numbness of limbs, lips and tongue, deafness, blurring of vision and mental derangement.
- A crippling deformity called Minamata disease due to consumption of fish captured from mercury contaminated Minamata Bay in Japan was detected in 1952.
- Water contaminated with cadmium can cause itai itai disease also called ouch-ouch disease (a painful disease of bones and joints) and cancer of lungs and liver.
- The compounds of lead cause anaemia, headache, loss of muscle power and bluish line around the gum.

Hazards of ground water pollution:

- Presence of excess nitrate in drinking water is dangerous for human health and may be fatal for infants.
- Excess nitrate in drinking water reacts with haemoglobin to form non-functional methaemoglobin and impairs oxygen transport. This condition is called methemoglobinemia or blue baby syndrome.
- Excess fluoride in drinking water causes neuro- muscular disorders, gastro-intestinal problems, teeth deformity, hardening of bones and stiff and painful joints (skeletal fluorosis).
- High concentration of fluoride ions is present in drinking water in 13 states of India. The maximum level of fluoride, which the human body can tolerate is 1.5 parts per million(mg/1 of water). Long term ingestion of fluoride ions causes fluorosis.

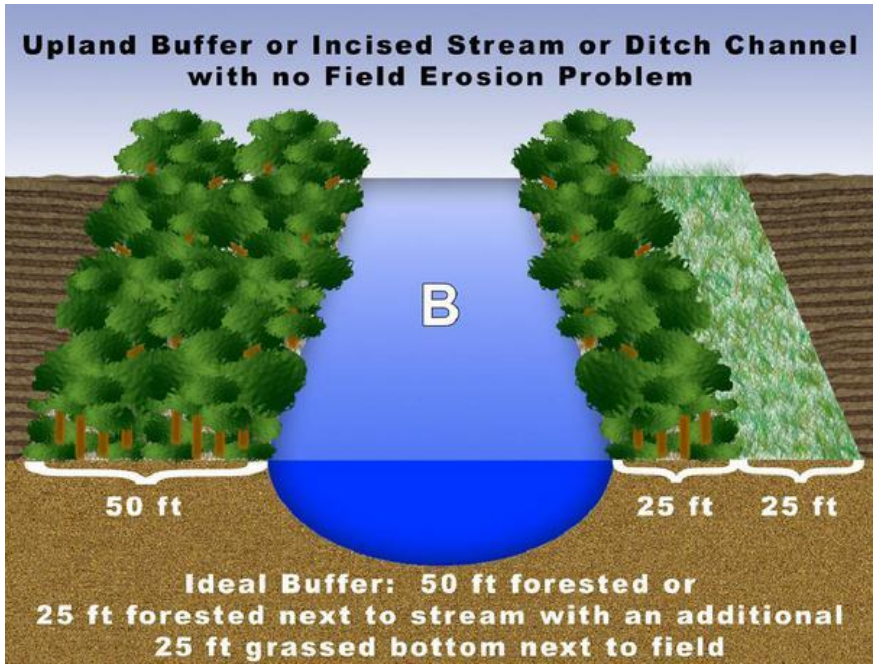
Hazards of ground water pollution:

- Over exploitation of ground water may lead to leaching of arsenic from soil and rock sources and contaminate ground water. Chronic exposure to arsenic causes black foot disease. It also causes Diarrhoea, Peripheral neuritis, hyperkeratosis and also lung and skin cancer.
- Arsenic contamination is a serious problem (in tube well dug areas) in the Ganges Delta, west Bengal causing serious arsenic poisoning to large numbers of people.
- A 2007 study found that over 137 million people in more than 70 countries are probably affected by arsenic poisoning of drinking water.

Control Measures

- Riparian buffers
- Recycle
- Treatment of sewage water and the industrial effluents should be done before releasing it water bodies.
- Hot water should be cooled before release from the power plants
- Domestic cleaning in tanks, streams and rivers, which supply drinking water, should be prohibited.
- Excessive use of fertilizers and pesticides should be avoided.
- Organic farming and efficient use of animal residues as fertilizers.
- Water hyacinth (an aquatic weed) can purify water by taking some toxic materials and several heavy metals from water.
- Oil spills in water can be cleaned with the help of bregoli a by-product of paper industry resembling saw dust, oil zapper, organisms.





Drinking Water standard



Fluoride:

Lack of fluoride in drinking water can lead to tooth decay.

It is important to add soluble fluoride in drinking water to make the concentration up to 1 ppm. However, the concentration should not cross 2 ppm.



Lead

The maximum permissible limit for the concentration of lead in drinking water is 50 parts per billion.



Sulfates & Nitrates

Sulfate concentration greater than 500 ppm in drinking water can cause the laxative effect.

Nitrate concentration in drinking water should not exceed 50 ppm.

Drinking Water standard

Metal	Maximum Concentration (ppm)
Zn	5.0
Fe	0.2
Mn	0.05
Cu	3.0
Al	0.2
Cd	0.005

Permissible Concentration of Few Metals in Drinking Water

INDIAN STANDARD SPECIFICATIONS FOR DRINKING WATER
IS: 10500

S.NO.	Parameter	Requirement desirable Limit	Remarks
1.	Colour	5	May be extended up to 50 if toxic substances are suspected
2.	Turbidity	10	May be relaxed up to 25 in the absence of alternate
3.	pH	6.5 to 8.5	May be relaxed up to 9.2 in the absence
4.	Total Hardness	300	May be extended up to 600
5.	Calcium as Ca	75	May be extended up to 200
6.	Magnesium as Mg	30	May be extended up to 100
7.	Copper as Cu	0.05	May be relaxed up to 1.5
8.	Iron	0.3	May be extended up to 1
9.	Manganese	0.1	May be extended up to 0.5
10.	Chlorides	250	May be extended up to 1000
11.	Sulphates	150	May be extended up to 400
12.	Nitrates	45	No relaxation
13.	Fluoride	0.6 to 1.2	If the limit is below 0.6 water should be rejected, Max. Limit is extended to 1.5
14.	Phenols	0.001	May be relaxed up to 0.002
15.	Mercury	0.001	No relaxation
16.	Cadmium	0.01	No relaxation
17.	Selenium	0.01	No relaxation
18.	Arsenic	0.05	No relaxation
19.	Cyanide	0.05	No relaxation
20.	Lead	0.1	No relaxation
21.	Zinc	5.0	May be extended up to 10.0
22.	Anionic detergents (MBAS)	0.2	May be relaxed up to 1
23.	Chromium as Cr^{*6}	0.05	No relaxation
24.	Poly nuclear aromatic Hydrocarbons	--	--
25.	Mineral Oil	0.01	May be relaxed up to 0.03
26.	Residual free Chlorine	0.2	Applicable only when water is chlorinated
27.	Pesticides	Absent	--
28.	Radio active	--	--

Wastewater

- Wastewater is water generated after the use of freshwater, raw water, drinking water or saline water in a variety of deliberate applications or processes.
- Wastewater is human changed waters with other physical, chemical and biological properties as natural waters.
- Classification of wastewaters by origin:
  - Household wastewater
  - Municipal wastewater
  - Precipitation (rain) wastewater
  - Industrial (production) wastewater

- Insufficient treatment of wastewater and faecal sludge spreads disease and is a driver of antimicrobial resistance.
- WHO produces global guidelines and tools to improve treatment and manage health risks at all steps of the chain where wastewater is used.



Industrial wastewater quantity

Types:

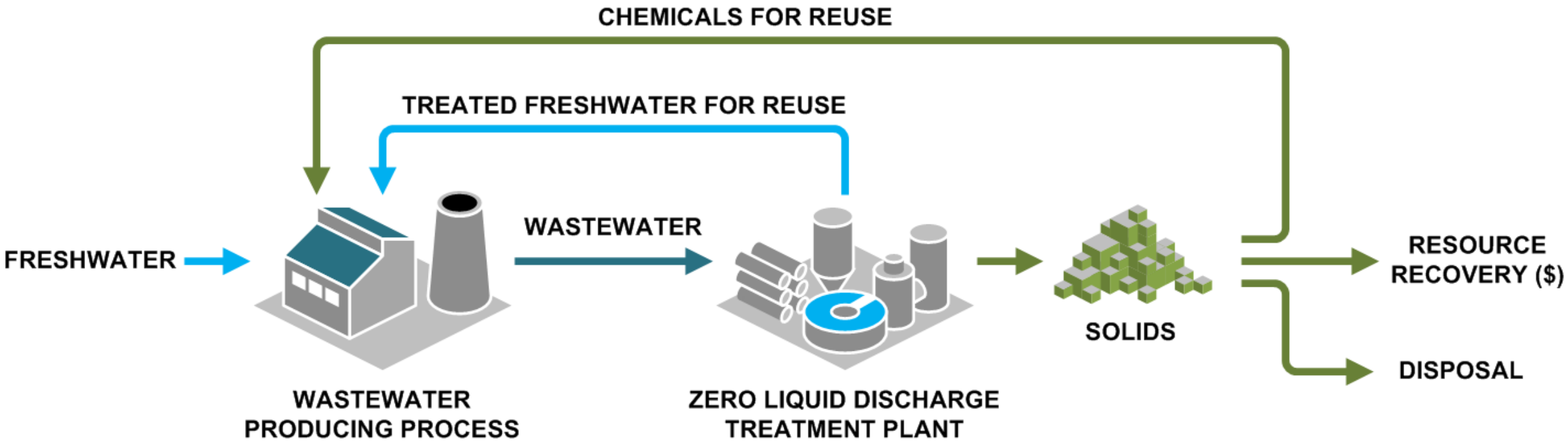
Relatively Polluted (used in HE),

Polluted (Chemicals),

Very Polluted (special treatment)

Wastewater amount by processing of the 1 unit (kg or L) of the product

- Meat 1 – 3L
- Canned fish 17-45 L
- Beer 2, 5 – 15 L
- Sugar 100 L
- Wool – 100 L
- Coal 20 -30 L
- Cellulose 80 – 150 L



Wastewater Treatment

- Physical – Flotation, Screen, Clarifier
- Chemical – Neutralization, flocculation & coagulation, Oxidation, Ion exchange, Ozonation, Disinfection
- Biological – Anaerobic digestion, aerobic sludge treatment, algae, etc.

Q & A?

- Most polluted river in the world?
- Most polluted river in India?
- Which state have water scarcity?
- Which state has more water pollution?
- Which state is top in households having access to safe drinking water?
- Which state has the cleanest water ?
- Among 122 countries in terms of the quality of drinking water, what is the rank of India?
- Which country has the cleanest water ?

Soil Pollution



Contents



Introduction



Sources



Impacts

Introduction

The soil is a resource for which there is no substitute.

Soil is essentially a natural body of mineral and organic constituents produced by solid material recycling.

The soil is the home for a large part of bacterial biodiversity and other microscopic and macroscopic living organisms.

Weathering of earth's crusts by different processes leads to the formation of soil that accumulates over the centuries.

Soil pollution is a major challenge for India.

Agriculture, with its allied sectors, is the largest source of livelihoods in India. 70 % of its rural households still depend primarily on agriculture for their livelihood, with 82 % of farmers being small and marginal.



Soil Pollution

Presence of a chemical or substance out of place and/or present at a higher-than-normal concentration that has adverse effects on any non-targeted organism.

Cannot be directly assessed or visually perceived, making it a hidden danger.

Soil contamination or soil pollution can occur either because of human activities or because of natural processes.

The soil contamination can occur due to the presence of chemicals such as pesticides, herbicides, ammonia, petroleum hydrocarbons, lead, nitrate, mercury, naphthalene, etc in an excess amount.

The primary cause of soil pollution is a lack of awareness in general people.

Many different human activities such as overuse of pesticides the soil will lose its fertility.

Introduction

India has a total land area of approximately 328 million hectares.

- Agricultural land : 43.6%
- Permanent pastures and meadows : 4.6%
- Cultural wastelands : 12.2%
- Forests : 10.7%
- Barren and uncultivable land : 8.4%
- Urban land : 5.3%
- No information available : 5.2%

Rapid industrial growth was also associated with surge in release of toxic effluents in the environment, including land and water bodies



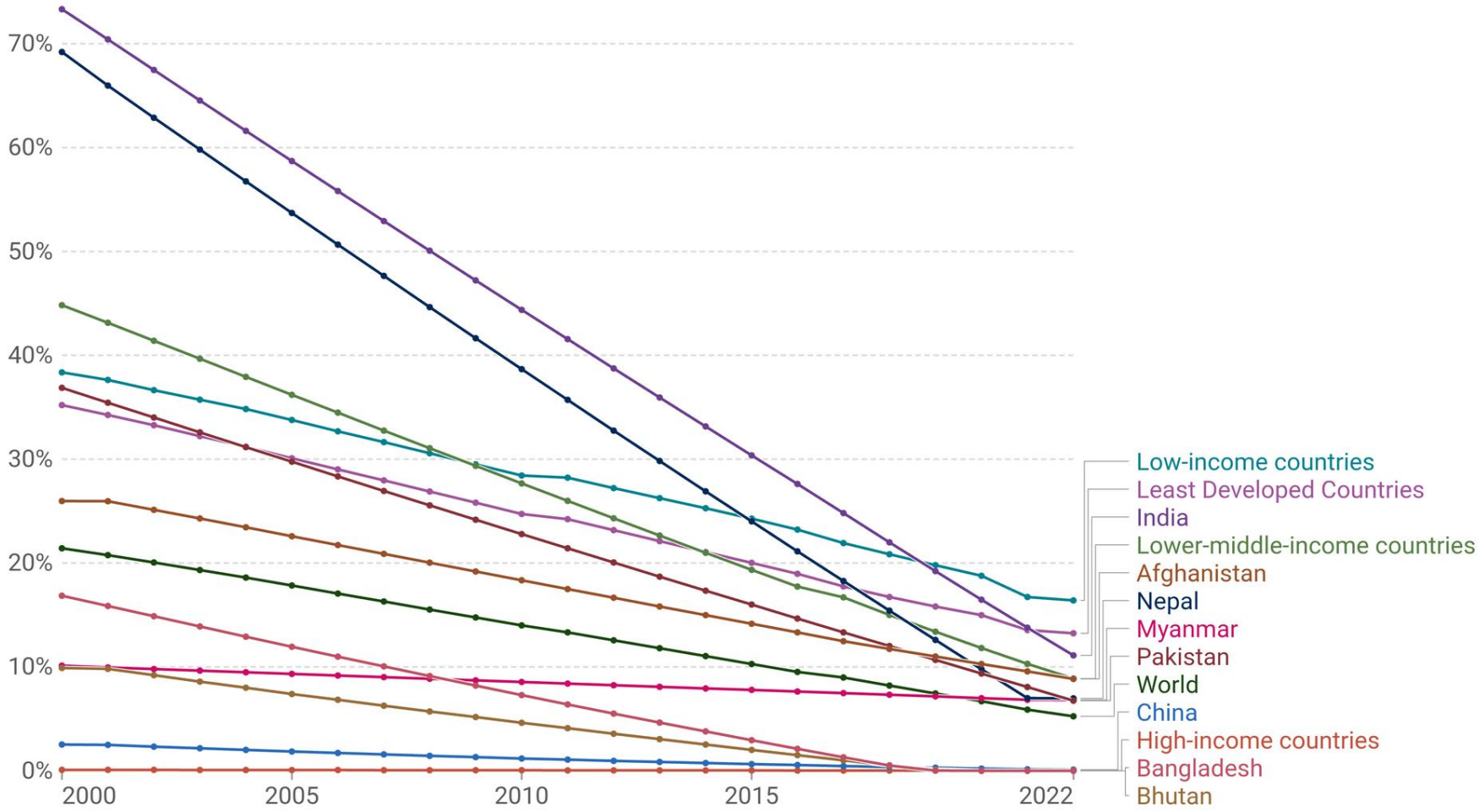
Swachh Bharat Abhiyan, or **Clean India Mission** is a country-wide campaign initiated by the Government of India on 2 October 2014 to eliminate open defecation and improve solid waste management and to create Open Defecation Free (ODF) villages.

Introduction

Share of people practicing open defecation

Open defecation refers to defecation in fields, forests, bushes, bodies of water, or other open spaces.

Our World in Data



Data source: WHO/UNICEF Joint Monitoring Programme (JMP) for Water Supply and Sanitation
OurWorldInData.org/sanitation | CC BY

Introduction

Yes, the Swachh Bharat Mission (SBM) has helped reduce soil pollution in India:

- **ODF villages**
  - According to a UNICEF study, villages that have achieved Open Defecation Free (ODF) status are less likely to have contaminated soil, water, and food than non-ODF villages.
- **Groundwater**
  - Groundwater in ODF villages is 12.7 times less likely to be contaminated with human waste than in non-ODF villages.
- **Piped water**
  - Piped water in ODF villages is 2.4 times less likely to be contaminated with human waste than in non-ODF villages.
- **Faecal contamination**
  - The risk of faecal contamination of household water in non-ODF villages is 2.68 times higher than in ODF villages.
- **Faecal contamination of food**
  - Faecal contamination of food in non-ODF villages is 1.48 times higher than in ODF villages.

To reduce the risk of contamination further, the SBM is focusing on solid and liquid waste management, as well as faecal sludge management.

Causes

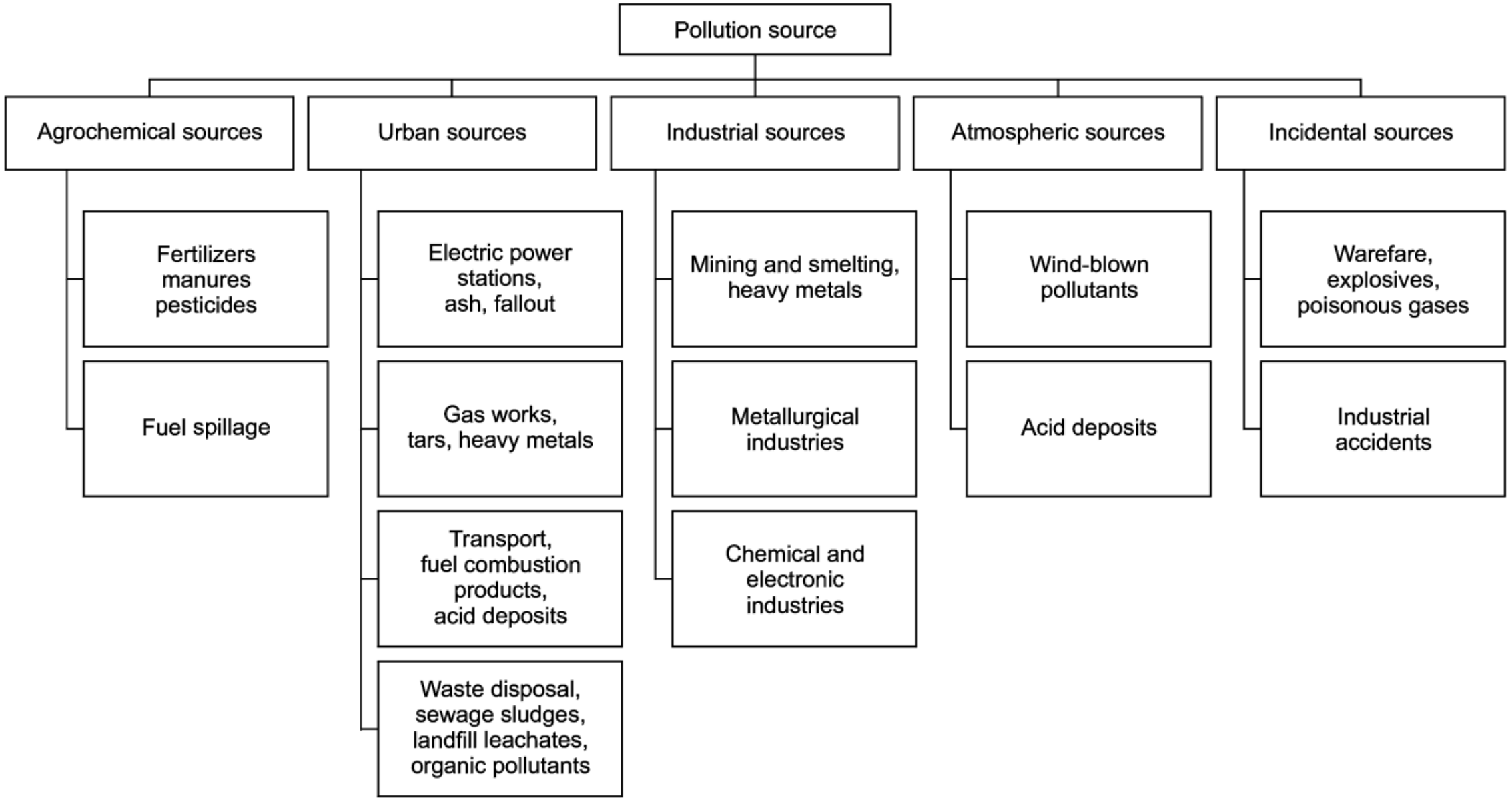
Natural

- Land Slide, Hurricanes, Natural disasters

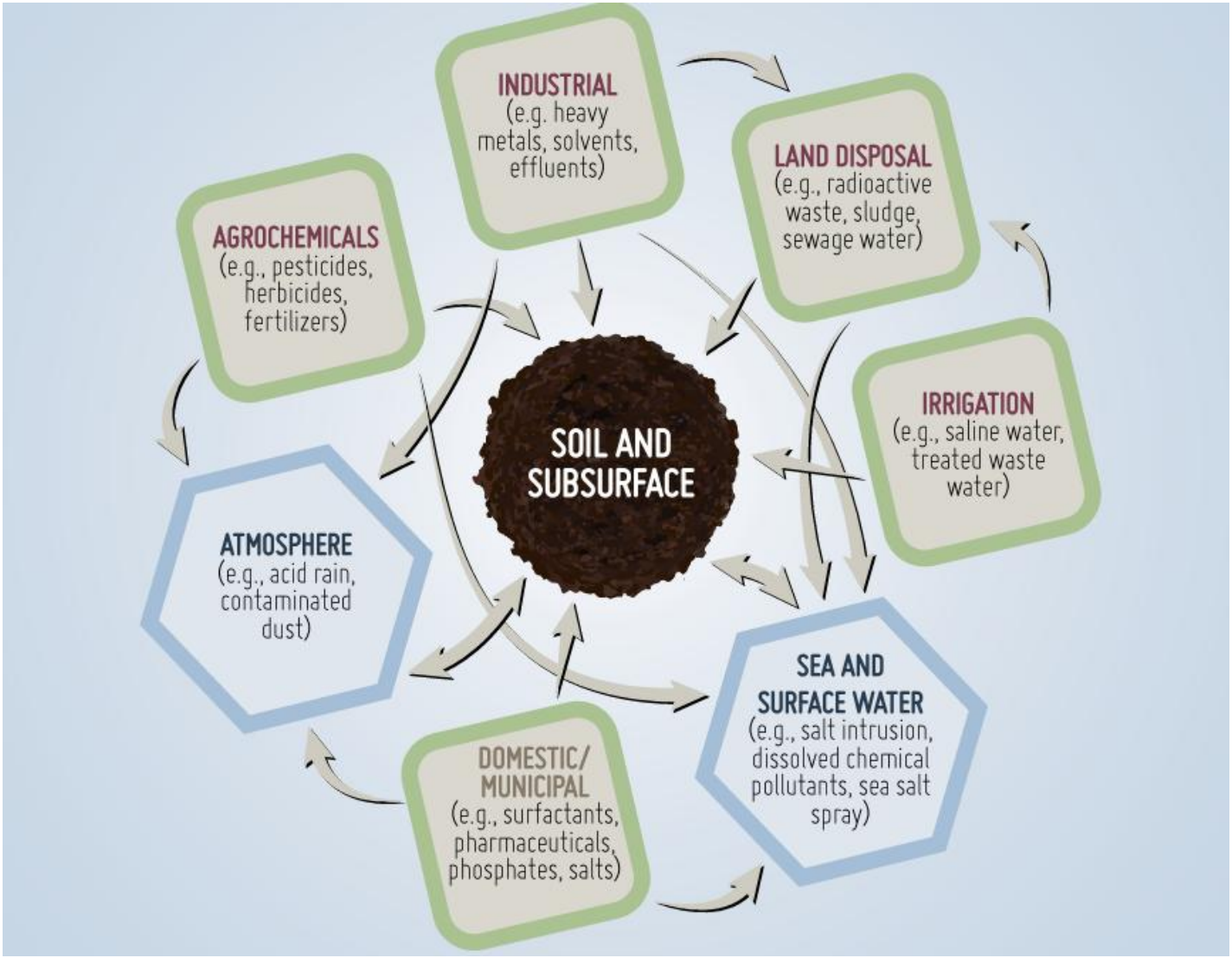
Man Made

- Urbanization, Industrial waste, Mining, Agricultural Waste, Domestic waste & Garbage, Radioactive waste

Sources of Soil Pollution



Sources of Soil Pollution



Sources : Agriculture



Sources of Soil Pollution

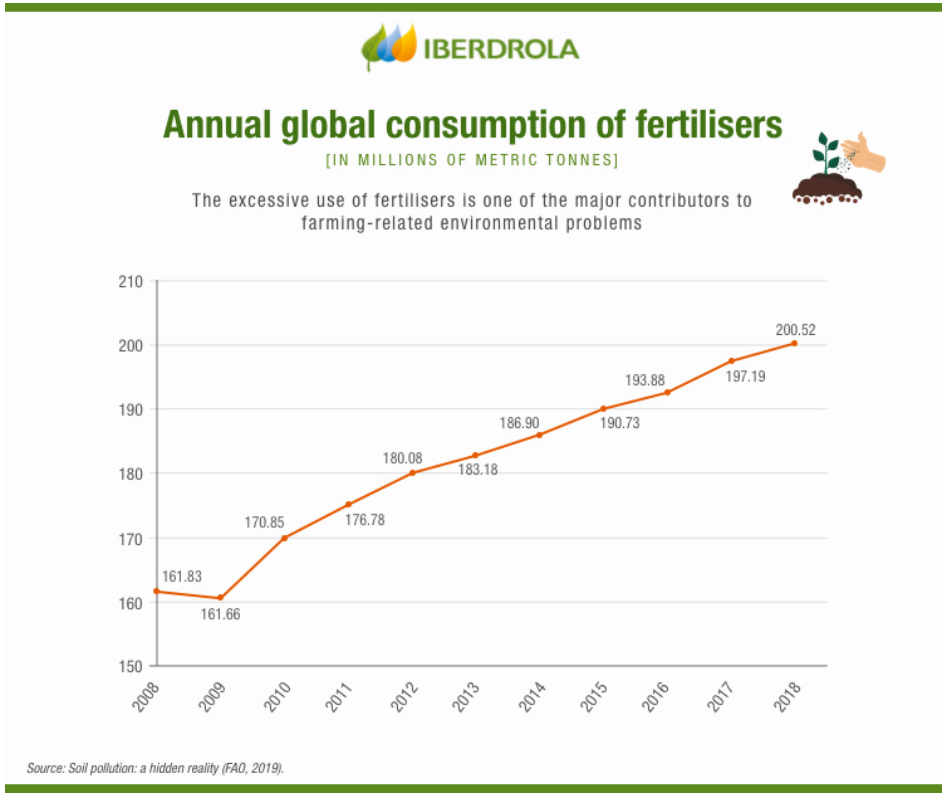
- Pollutants of Agrochemical Sources

  (fertilisers, manure, and pesticides)

  - Examples of these are the introduction of arsenic, cadmium, manganese, uranium, vanadium and zinc by some phosphate fertilizers, or soil contamination with zinc, arsenic and copper when poultry or pig manure materials are used.

Pesticides are generally classified into the following groups

- Insecticides. These are chemical compounds used to kill insects, whether specifically for a particular type or generally for a variety of insects.
- Herbicides. Chemicals used to combat or suppress the growth of all or certain types of plants.
- Fungicides. Chemicals used to kill or suppress the growth of all kinds or of a certain type of fungus.

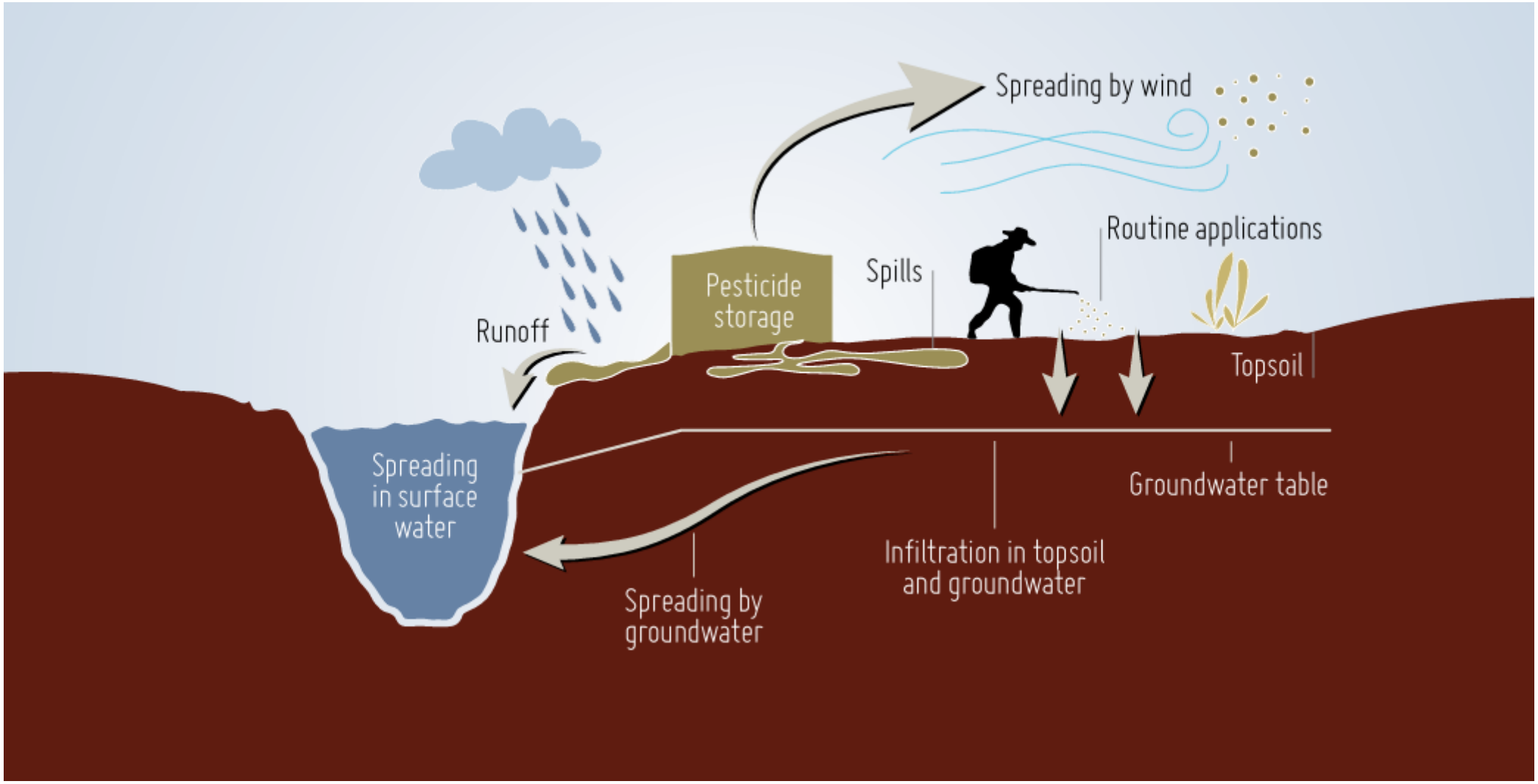


Sources of Soil Pollution

Insecticides :

- Organophosphorus compounds: tetraethyl pyrophosphate – TEPP, Sarin
- Organochlorines: DDT (dichlorodiphenyl trichloroethane) family, the BHC (benzene hexachloride) family and the cyclodiene family
- Carbamates: These are derivatives of carbamic acid NH_2–COOH
- Pyrethroids: Pyrethroids were originally quite effective natural pesticides, extracted from Chrysanthemum cineraria folium – a plant that was for centuries grown specially in Persia to obtain these substances. Nowadays the main producers of natural pyrethrum are Kenya and Tanzania.

Sources of Soil Pollution



Sources of Soil Pollution

❑Pollutants of Urban Sources

The main sources of urban soil pollution, however, are power generation emissions, releases from transport means and waste disposal.

- Power Generation Emissions
- Soil Pollution through Transport Activities
- Soil Pollution by Waste and Sewage Sludge: *These waste disposed of by humans contain chemical waste from residential areas. Moreover, leaking of sewerage system can also affect soil quality and cause soil pollution by changing the chemical composition of the soil*

Sources of Soil Pollution

Industrial Sources

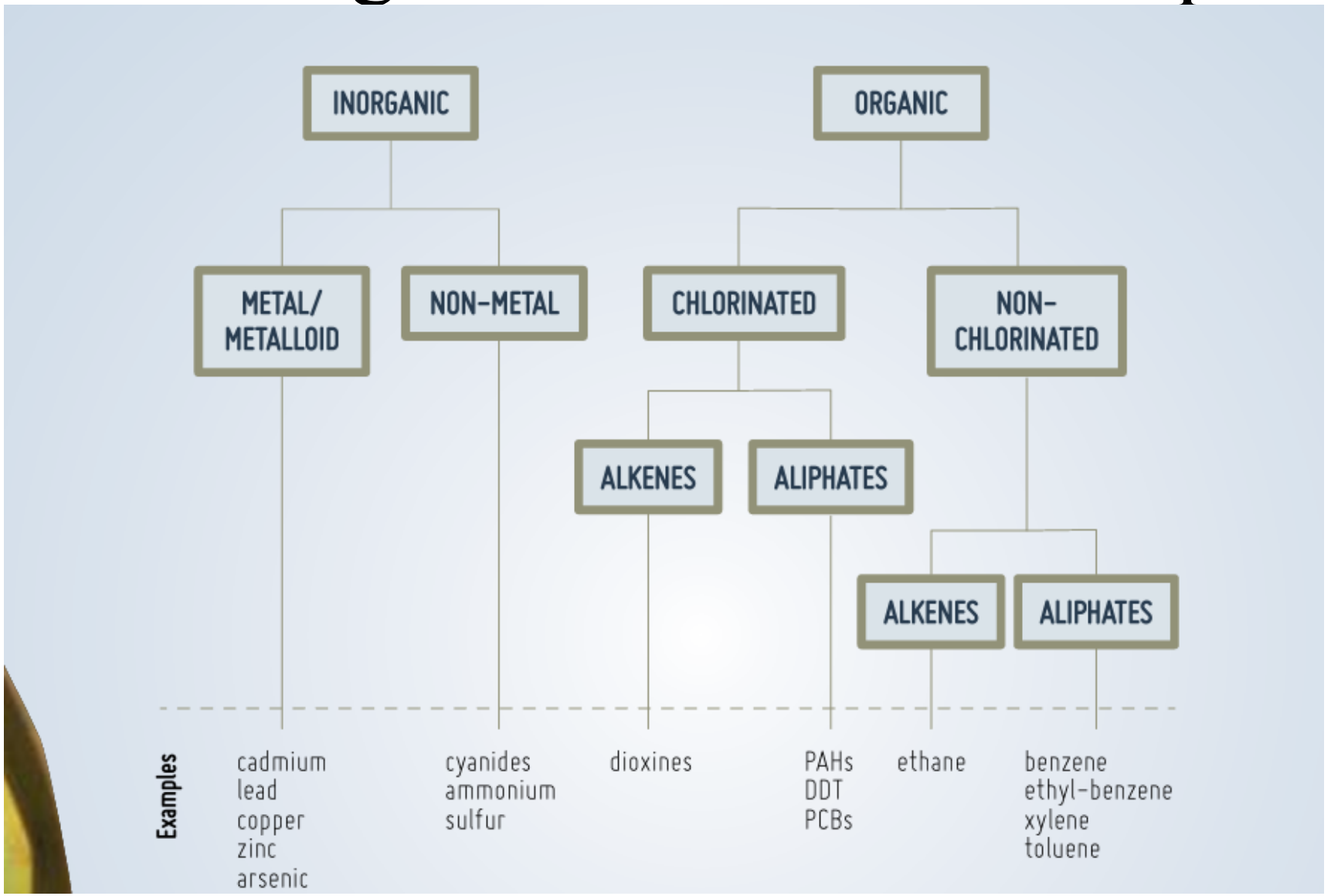
Pulp and paper mills, chemical fertilizers, oil refineries, sugar factories, tanneries, textiles, steel, distilleries, fertilizers, pesticides, coal and mineral mining industries, drugs, glass, cement, petroleum and engineering industries etc.

Industrial waste includes chemicals such as mercury, lead, copper, zinc, cadmium, cyanides, thiocynates, chromates, acids, alkalies, organic substances etc.

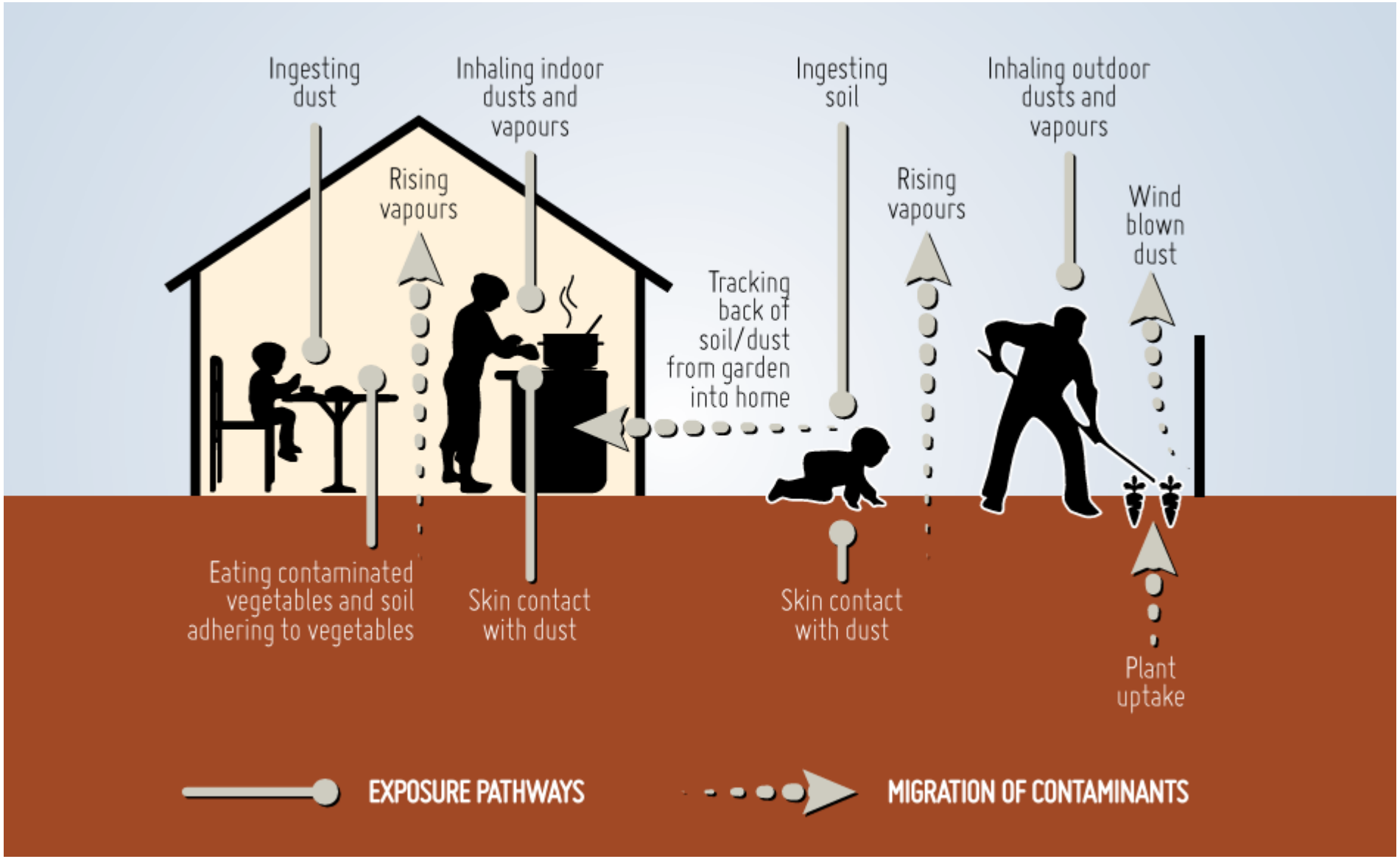
Sources of Soil Pollution

- Radioactive wastes: - Radioactive elements from mining and nuclear power plants, find their way into water and then into the soil.
- Nuclear reactors produce waste containing Ruthenium-106, Iodine-131, Barium140, Cesium- 144 and Lanthanum-140 along with primary nuclides Sr-90 with a half life 28 years and Cs- 137 with a half life 30 years.
- Rain-water carries Sr-90 and Cs-137 to be deposited on the soil where they are held firmly with the soil particles by electrostatic forces.
- All the radio nuclides deposited on the soil emit gamma radiations.

Systematic categorization of the main pollutants



Possible exposure pathways of soil contamination in a residential scenario



Effects of Soil Pollution

- Agriculture –
  - Reduced soil fertility
  - Reduced nitrogen fixation
  - Increased erosion
  - Loss of soil and nutrients
  - Reduced crop yield
  - Increased salinity
  - Deposition of silt in tanks and reservoirs

Effects of Soil Pollution

Health

- Dangerous chemicals entering underground water
- Bio magnification
- Release of pollutant gases
- Release of radioactive rays causing health problems

Environment

- Reduced vegetation
- Ecological imbalance
- Imbalance in soil fauna and flora

Preventive measures / Control

- Use of correct farming techniques
- Recycling of Waste before disposal
- Proper disposal method of household and industrial waste
- Use of organic fertilizers instead of chemical fertilizers and pesticides
- Community education and awareness
- Proper maintenance of sewage system

Thank you